THEFT-DETERRENT SYSTEM

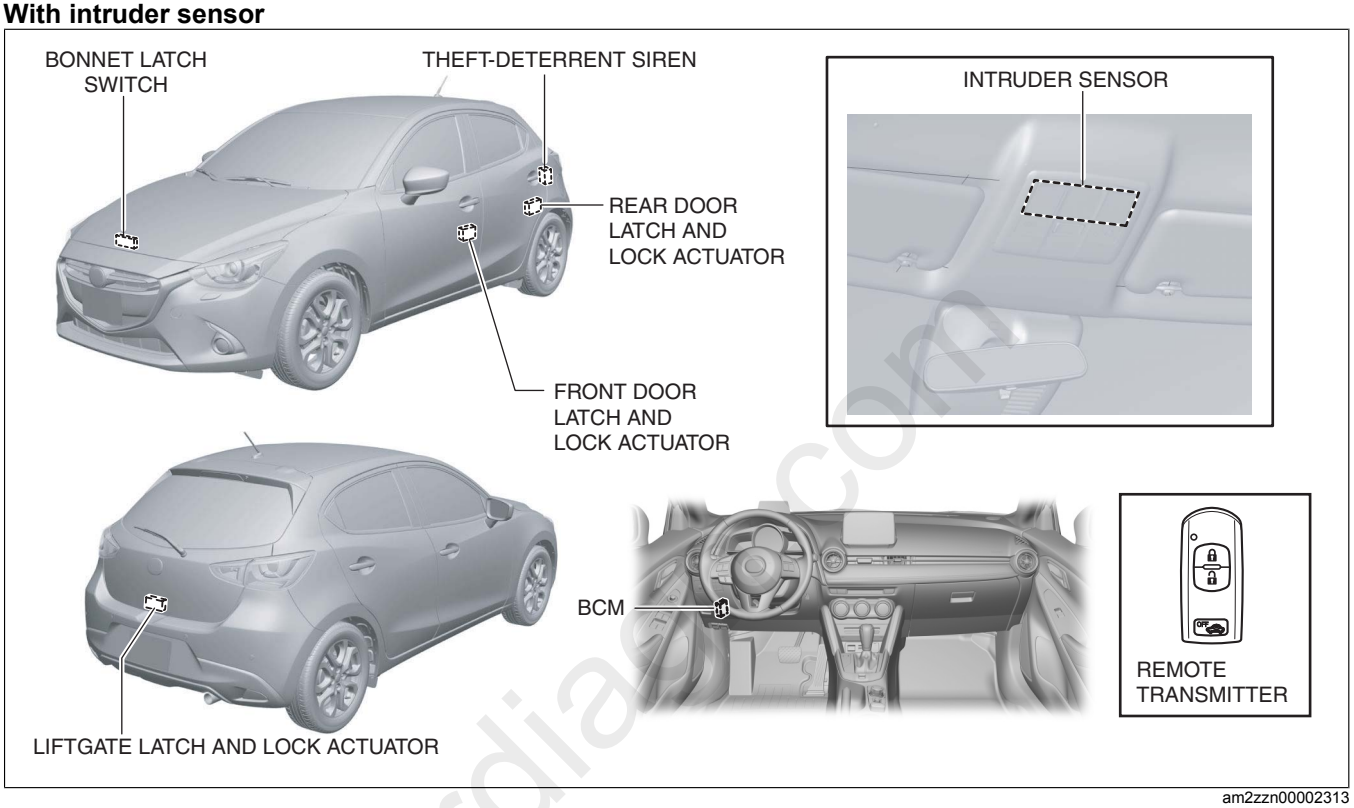
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Outline

- The theft-deterrent system operates the theft-deterrent siren or theft-deterrent horn and hazard warning lights if a door, liftgate/trunk lid or the bonnet is improperly opened.
- An answer-back function has been adopted which allows visual verification of the theft-deterrent system operation.
- Signal input/output of the theft-deterrent system is controlled by the BCM.

Structural View

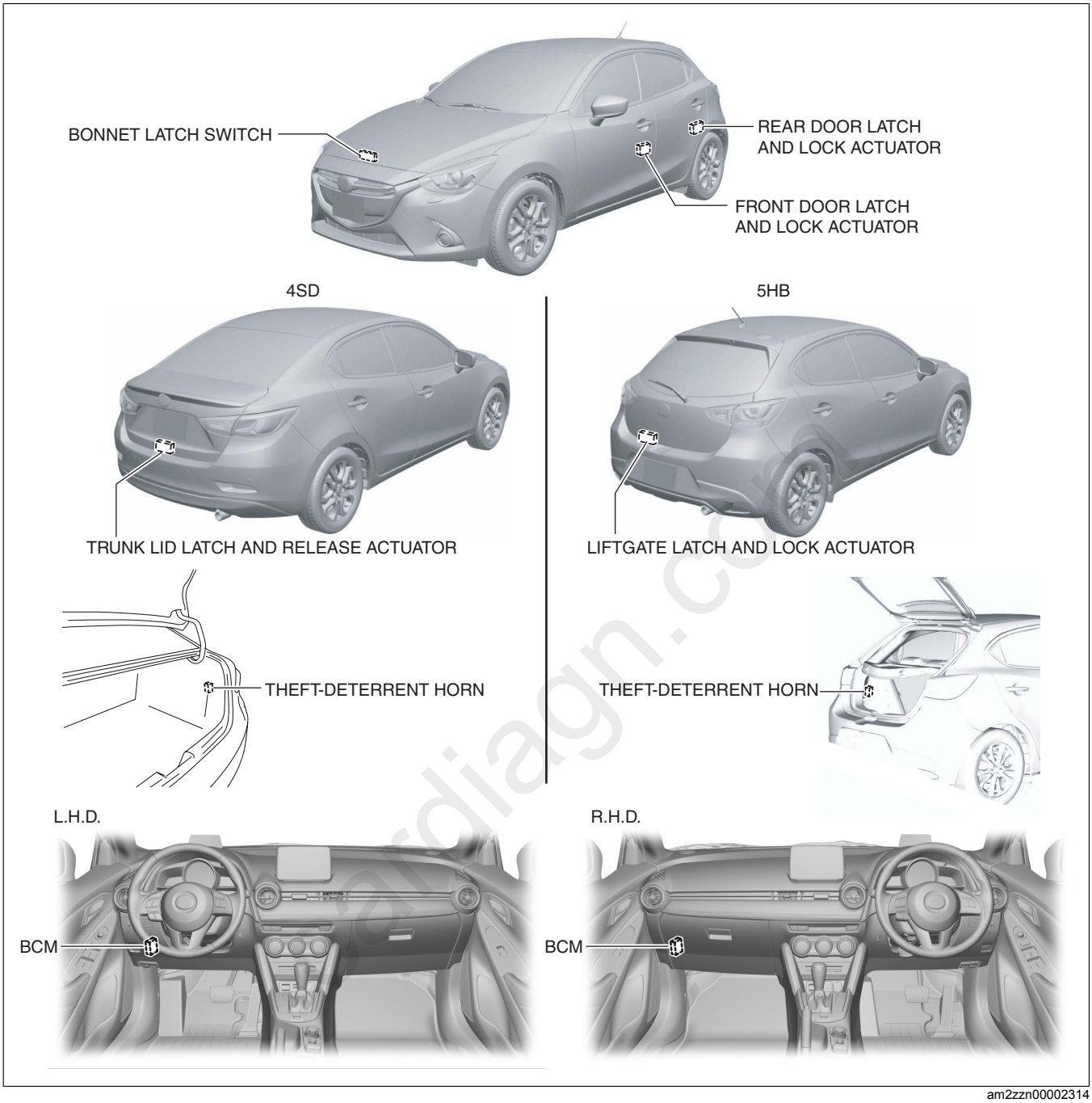
With intruder sensor



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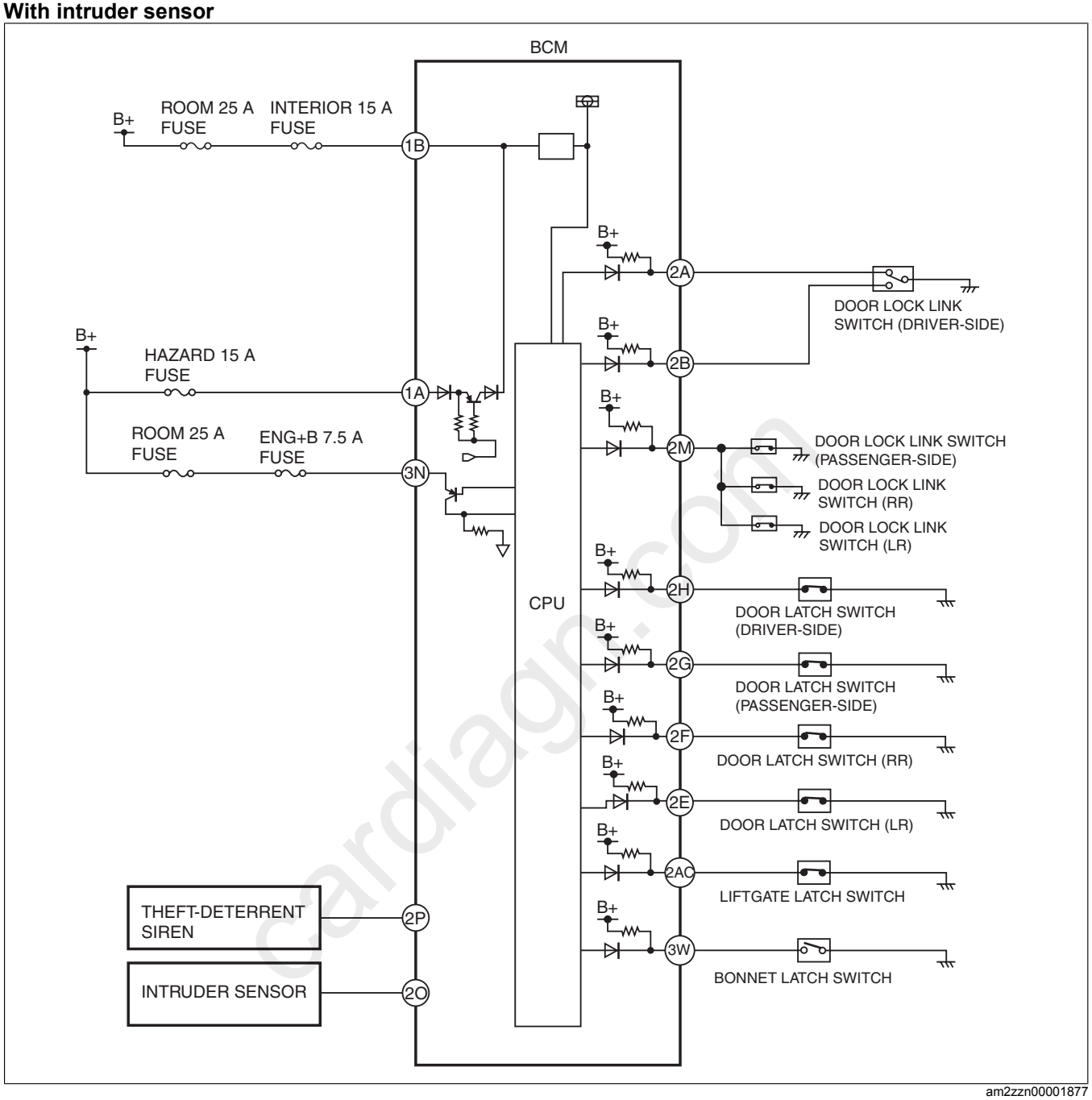
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Without intruder sensor



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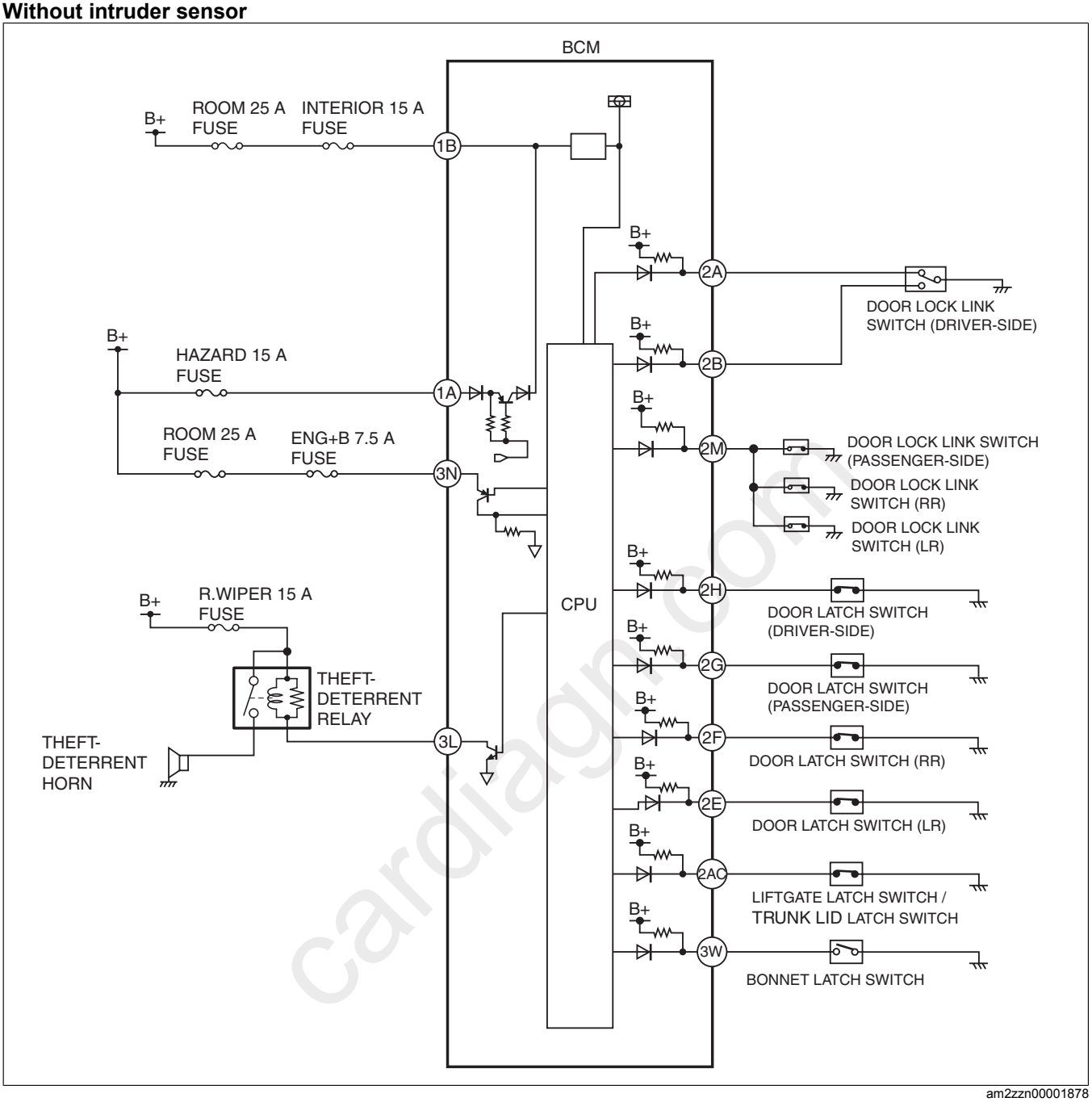
System Wiring Diagram
With intruder sensor



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Without intruder sensor



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Function

Answer back function

- When a locking operation is performed using the remote transmitter is in stand-by preparation mode, stand-by mode, the hazard warning lights flash once.
- When a unlocking operation is performed using the remote transmitter is in initial mode, the hazard warning lights flash twice.

Intruder sensor cancel function

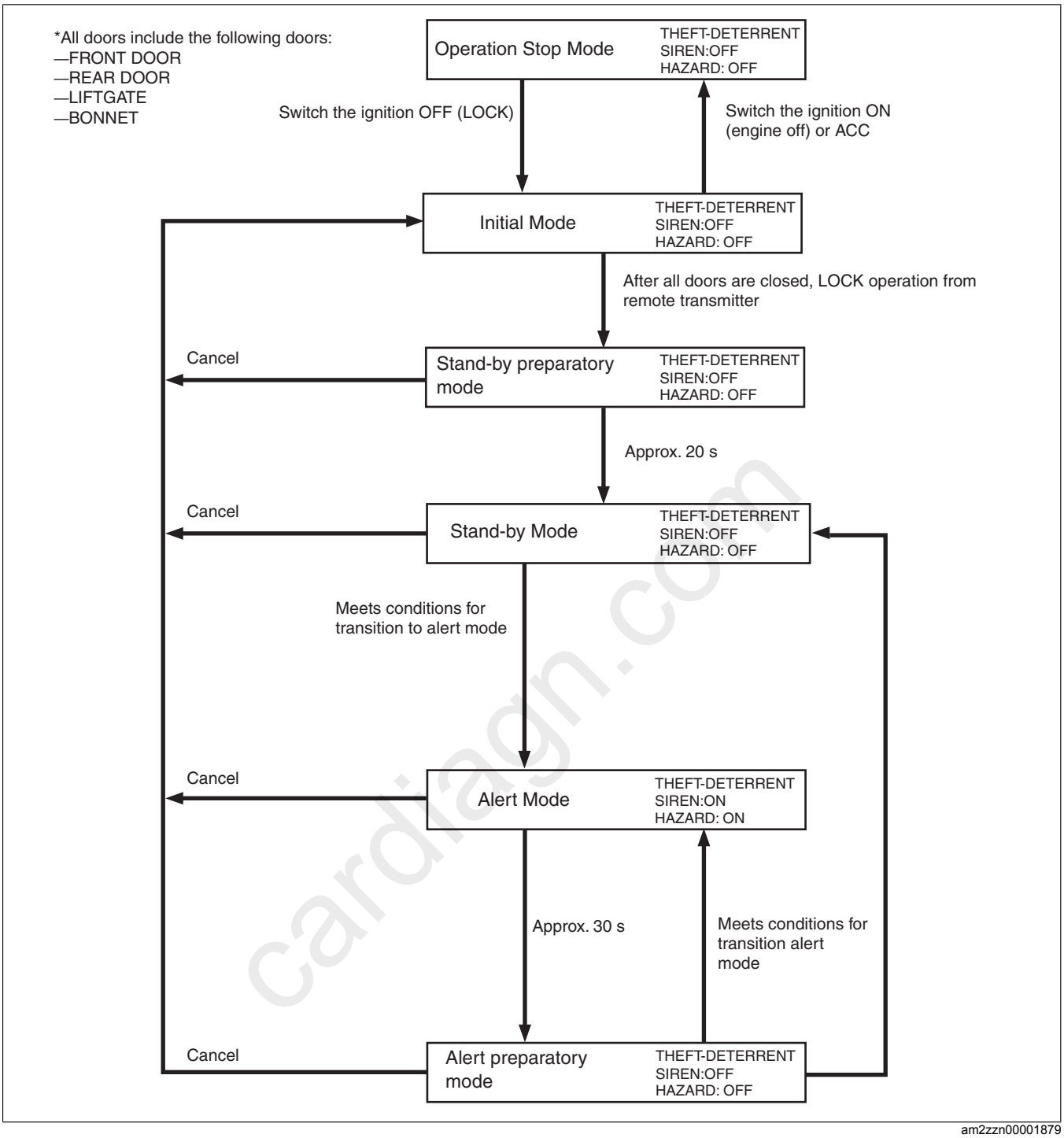
- An intruder sensor cancel button has been installed on the remote transmitter. If the intruder sensor cancel button is on while the theft-deterrent system is stand-by preparatory mode, intruder sensor ultrasonic wave output is cut, and movement detection is stopped.

Operation (With Intruder Sensor)

Mode transition

- The BCM switches the theft-deterrent system mode based on the ignition switch and door conditions.

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Operation stop mode

- In this mode, all theft-deterrent system operations are stopped.
- The system changes to the operation stop mode when any of the following conditions are met.
  — Switch the ignition to ACC or ON (engine off).

Initial mode

- This mode is the preparatory mode before changing to the stand-by mode. The system does not change directly to the alert mode from this mode.

Transition condition to the initial mode

- The system changes from the operation stop mode to the initial mode if all of the following conditions are met.
  — Switch the ignition off
- The system transitions from stand-by preparatory mode, stand-by mode, alert mode, and alert preparatory mode to the initial mode if the cancel conditions in each mode are met.

Transition condition to the stand-by preparatory mode

- The system changes to the stand-by preparatory mode if any of the following conditions are met.

— A lock operation was performed using the remote transmitter lock button with all the doors, liftgate, and bonnet closed.

Stand-by preparatory mode
- This is mode prior to the stand-by mode.
- If the cancel conditions are met, the system changes to the initial mode.

Transition condition to the stand-by mode
- If the stand-by preparatory mode continues for **approx. 20 s**, the system changes to the stand-by mode.

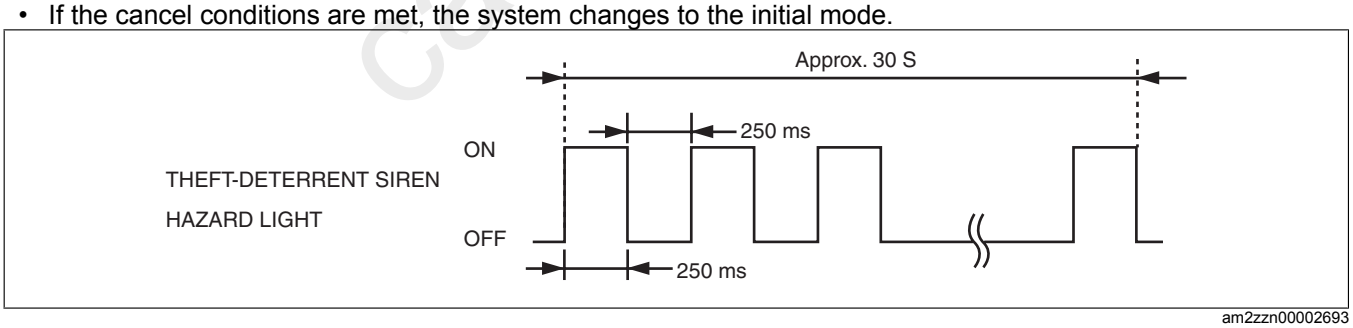
Cancel conditions
- If any of the following conditions are met, the stand-by preparatory mode is canceled and the system changes to the initial mode.
  — Any door latch switch on (door is open), or liftgate latch switch on (liftgate is open), or bonnet latch switch off (bonnet is open)
  — Any door lock-link switch UNLOCK
  — Switch the ignition to ACC or ON (engine off).
  — Unlock operation using remote transmitter
  — Open the liftgate using the liftgate opener switch (With advanced keyless entry system)

Stand-by mode
- The mode monitors for unauthorized door opening/closing **approx. 20 s** after the doors are locked. If any unauthorized operation is detected, the system changes to alert mode.
- If the cancel conditions are met, the system changes to the initial mode.

Transition condition to the alert mode
- If any of the following conditions are met, the system changes to the alert mode.
  — Forced ignition ON or ignition is switched ON without transmitter authorization
  — Door latch switch on (door is open)
  — Liftgate latch switch on (liftgate is open)
  — Bonnet latch switch off (bonnet is open)
  — Door lock-link switch UNLOCK
  — Intrusion into vehicle is detected by intruder sensor

Cancel condition
- If any of the following conditions are met, the stand-by mode is canceled and the system changes to the initial mode.
  — Unlock operation using remote transmitter
  — Switch the ignition to ON (engine off), and engine start
  — Open the liftgate using the liftgate opener switch (With advanced keyless entry system)

Alert mode
- In this mode, the theft-deterrent siren and hazard warning lights are operated to alert the surrounding area.
- When the alarm period (approx. 30 s) has elapsed, the system changes to the alert preparatory mode.
- If the cancel conditions are met, the system changes to the initial mode.



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Transition condition to the alert preparatory mode
- When the alarm period (**approx. 30 s**) has elapsed, the system changes to the alert preparatory mode.
- The alarm period is not extended even if a transition condition to alert mode is met again during an alarm period.

Cancel condition
- If any of the following conditions are met, the alert mode is canceled, and the system changes to the initial mode.
  — Unlock operation using remote transmitter
  — Switch the ignition to ON (engine off), and engine start
  — Open the liftgate using the liftgate opener switch (With advanced keyless entry system)

Alert preparatory mode
- In the alert preparatory mode, the theft-deterrent siren and hazard warning light are not operated, however, the alarm triggers immediately if the condition is changed.
- If the cancel conditions are met, the system changes to the initial mode.

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Transition condition to alert mode

- The system changes to the alert mode if any of the following conditions are met.
  - Forced ignition ON or ignition is switched ON without transmitter authorization
  - Door latch switch on (door is open)
  - Liftgate latch switch on (liftgate is open)
  - Bonnet latch switch off (bonnet is open)
  - Door lock-link switch UNLOCK
  - Intrusion into vehicle is detected by intruder sensor

Transition condition to the stand-by mode

- If there is no transition to cancel or alert mode for a period of 5 s after warning stand-by mode is established, the system switches to stand-by mode.

Cancel condition

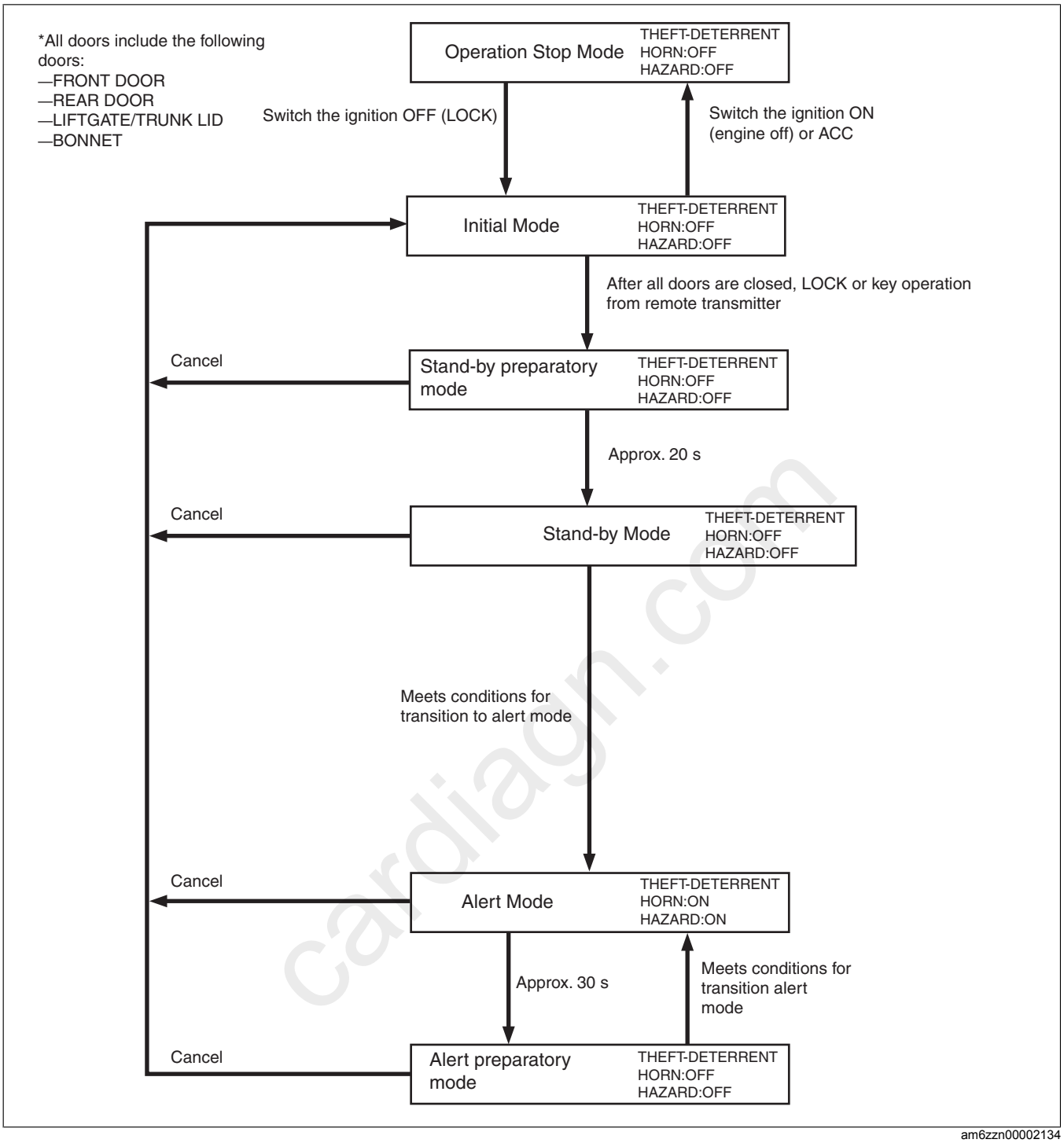
- If any of the following conditions are met, the alert preparatory mode is canceled and the system changes to the initial mode.
  - Unlock operation using remote transmitter
  - Switch the ignition ON (engine off), and engine start
  - Open the liftgate using the liftgate opener switch (With advanced keyless entry system)

Operation (Without Intruder Sensor)

Mode transition

- The BCM switches the theft-deterrent system mode based on the ignition switch and door conditions.

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Operation stop mode

- In this mode, all theft-deterrent system operations are stopped.
- The system changes to the operation stop mode when any of the following conditions are met.
  — Switch the ignition to ACC or ON (engine off).

Initial mode

- This mode is the preparatory mode before changing to the stand-by mode. The system does not change directly to the alert mode from this mode.

Transition condition to the initial mode

- The system changes from the operation stop mode to the initial mode if all of the following conditions are met.
  — Switch the ignition off
- The system transitions from stand-by preparatory mode, stand-by mode, alert mode, and alert preparatory mode to the initial mode if the cancel conditions in each mode are met.

Transition condition to the stand-by preparatory mode

- The system changes to the stand-by preparatory mode if any of the following conditions are met.

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— A lock operation was performed using the remote transmitter lock button or key with all the doors, liftgate/trunk lid, and bonnet closed.

Stand-by preparatory mode
- This is mode prior to the stand-by mode.
- If the cancel conditions are met, the system changes to the initial mode.

Transition condition to the stand-by mode
- If the stand-by preparatory mode continues for **approx. 20 s**, the system changes to the stand-by mode.

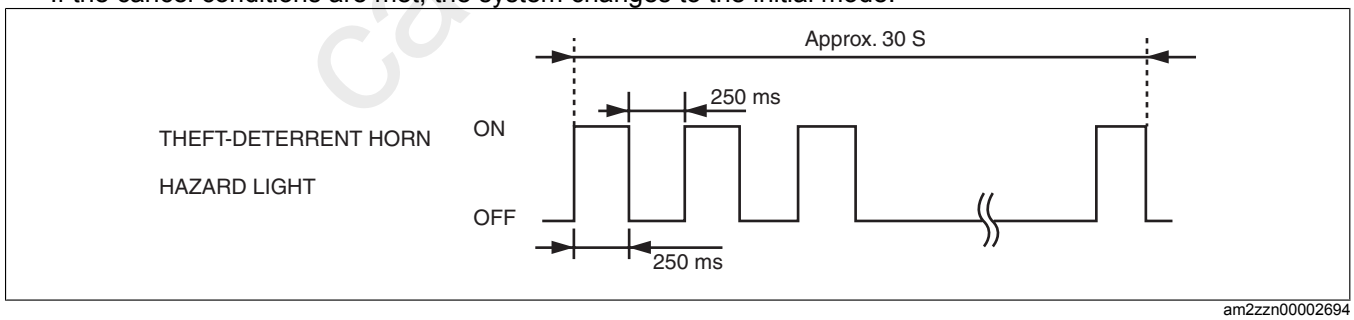
Cancel conditions
- If any of the following conditions are met, the stand-by preparation mode is canceled and the system changes to the initial mode.
  - — Any front door latch switch on (door is open), or any rear door latch switch on (rear door is open), or liftgate/trunk lid latch switch on (liftgate/trunk lid is open), or bonnet latch switch off (bonnet is open)
  - — Any door lock-link switch UNLOCK
  - — Switch the ignition to ACC or ON (engine off)
  - — Unlock operation using remote transmitter
  - — Open the liftgate/trunk lid using the liftgate/trunk lid opener switch (With advanced keyless entry system)

Stand-by mode
- The system starts monitoring **approx. 20 s** after the doors are locked for unauthorized door opening/closing. If any unauthorized operation is detected, the system changes to the alert mode.
- If the cancel conditions are met, the system changes to the initial mode.

Transition condition to the alert mode
- If any of the following conditions are met, the system changes to the alert mode.
  - — Switch the ignition ON (engine off)
  - — Door latch switch on (door is open)
  - — Liftgate/trunk lid latch switch on (liftgate/trunk lid is open)
  - — Bonnet latch switch off (bonnet is open)
  - — Door lock-link switch UNLOCK
  - — Unlock operation using front door key cylinder

Cancel condition
- If any of the following conditions are met, the stand-by mode is canceled and the system changes to the initial mode.
  - — Unlock operation using remote transmitter
  - — Switch the ignition to ACC or ON (engine off).

Alert mode
- In this mode, the theft-deterrent horn and hazard warning lights are operated to alert the surrounding area.
- When the alarm period (**approx. 30 s**) has elapsed, the system changes to the alert preparatory mode.
- If the cancel conditions are met, the system changes to the initial mode.



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Transition condition to the alert preparatory mode
- When the alarm period (**approx. 30 s**) has elapsed, the system changes to the alert preparatory mode.
- The alarm period is not extended even if a transition condition to alert mode is met again during an alarm period.

Cancel condition
- If any of the following conditions are met, the alert mode is canceled, and the system changes to the initial mode.
  - — Unlock operation using remote transmitter
  - — Switch the ignition ON (engine off), and engine start
  - — Open the liftgate/trunk lid using the liftgate/trunk lid opener switch (With advanced keyless entry system)

Alert preparatory mode
- In the alert preparatory mode, the theft-deterrent horn and hazard warning light are not operated, however, the alarm triggers immediately if the condition is changed.
- If the cancel conditions are met, the system changes to the initial mode.

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Transition condition to alert mode

- The system changes to the alert mode if any of the following conditions are met.
  - The ignition is switched from ON to off, or from off to ON
  - The door latch switch is turned from on (door is open) to off (door is closed), or from off (door is closed) to on (door is open)
  - The liftgate/trunk lid latch switch is turned from on (liftgate/trunk lid is open) to off (liftgate/trunk lid is closed), or from off (liftgate/trunk lid is closed) to on (liftgate/trunk lid is open)
  - The bonnet latch switch is turned from off (bonnet is open) to on (bonnet is closed) or from on (bonnet is closed) to off (bonnet is open)
  - The door lock-link switch is locked from unlock, or unlocked from lock
  - The front door key cylinder is locked from unlock, or unlocked from lock

Cancel condition

- If any of the following conditions are met, the alert preparatory mode is canceled and the system changes to the initial mode.
  - Unlock operation using remote transmitter
  - Switch the ignition ON, and engine start
  - Open the liftgate/trunk lid using the liftgate/trunk lid opener switch (With advanced keyless entry system)

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